

Ecommerce Customer Churn Analysis

Project Objective

The objective of this project is to **analyze customer churn patterns in the e-commerce domain** using historical transactional data. By examining customer attributes such as tenure, preferred payment modes, satisfaction scores, complaints, and purchase behavior, the project aims to uncover the key drivers of churn. The ultimate goal is to equip e-commerce enterprises with **actionable insights** that enable targeted retention strategies, strengthen customer relationships, and ensure long-term business viability in a competitive marketplace.

Problem Statement

Customer churn is a major challenge in e-commerce, reducing lifetime value, raising acquisition costs, and weakening loyalty. Attrition often stems from poor onboarding, dissatisfaction, payment trust issues, or delivery gaps. Without structured analysis, businesses risk losing valuable customers. This project builds a data-driven churn model to identify vulnerable segments, link churn to behavioral and demographic factors, and provide actionable recommendations to reduce attrition.

Scope of the Project

The scope of this project covers the **end-to-end churn analysis pipeline** for an e-commerce dataset. It includes:

- **Data Cleaning:** Handling missing values, imputing mean/mode, and removing outliers to ensure dataset integrity.
- **Data Transformation:** Standardizing categorical values, renaming columns for clarity, creating derived fields (ComplaintReceived, ChurnStatus), and dropping redundant columns.
- **Exploratory Analysis:** Running SQL queries to measure churn rates across dimensions such as payment mode, tenure, satisfaction, complaints, and city tier.
- **Segmentation:** Identifying high-risk customer segments based on combinations of attributes (e.g., payment mode + order category + gender).
- **Dashboard Integration:** Creating SQL views (vw_churn_summary, vw_churn_by_payment, etc.) for seamless integration with BI tools like Power BI and Tableau.
- **Insight Synthesis:** Translating query outputs into executive-ready insights and recommendations for retention strategies.

Methodology

The methodology outlines the structured, step-by-step approach used to analyze customer churn in the e-commerce dataset. It ensures data integrity, actionable insights, and dashboard-ready outputs.

1. Data Collection

- Source: Historical transactional and behavioral data from the customer_churn table.
- Attributes include tenure, payment modes, satisfaction scores, complaints, order categories, and demographic details.

2. Data Cleaning

- **Missing Values:**
 - Numeric columns (WarehouseToHome, HoursSpentOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder) imputed with mean values (rounded).
 - Categorical columns (Tenure, CouponUsed, OrderCount) imputed with mode values.
- **Outliers:** Rows with WarehouseToHome > 100 removed to maintain realistic delivery distances.
- **Validation:** Post-cleaning checks ensured no remaining NULLs.

3. Data Transformation

- **Standardization:**
 - Replaced inconsistent values (Phone → Mobile Phone, Mobile → Mobile Phone, COD → Cash on Delivery, CC → Credit Card).
- **Column Renaming:**
 - PreferredOrderCat → PreferredOrderCat
 - HourSpendOnApp → HoursSpentOnApp
- **Derived Columns:**
 - ComplaintReceived → "Yes" if Complain = 1, else "No".
 - ChurnStatus → "Churned" if Churn = 1, else "Active".
- **Column Dropping:** Removed redundant fields (Churn, Complain) after transformation.

4. Exploratory Analysis

- SQL queries designed to measure churn across dimensions:
 - **Overall churn rate**
 - **Payment mode**
 - **Tenure (customer lifecycle stage)**
 - **Satisfaction score**
 - **ComplaintReceived flag**
 - **City tier (geographic segmentation)**
 - **High-risk segments (payment mode + order category + gender)**

5. Dashboard Integration

- Created modular SQL views (vw_churn_summary, vw_churn_by_payment, vw_churn_by_tenure, etc.) for direct BI tool integration.
- Views ensure consistency, reusability, and plug-and-play capability in Power BI/Tableau dashboards.

6. Insight Synthesis

- Translated query outputs into executive insights:
 - High churn in COD customers → trust/delivery issues.
 - Low satisfaction scores → strong churn predictor.
 - ComplaintReceived → direct churn correlation.
 - Tier 2/3 cities → regional logistics challenges.
- Identified high-risk customer segments for targeted retention campaigns.

7. Recommendations & Action Plan

- Lifecycle-based engagement (onboarding vs. loyalty).
- Promote digital payments with incentives.
- Strengthen complaint resolution processes.
- Invest in Tier 2/3 logistics and localized marketing.
- Segment-specific retention strategies (e.g., fashion + COD customers).

Insights from Each Query

1. Overall Churn Rate

- **What it shows:** The percentage of customers who have churned vs. stayed active.
- **Insight:** A high churn rate signals urgent retention issues. If churn is >20%, you need immediate interventions like loyalty programs, personalized offers, or better onboarding.

2. Churn by Payment Mode

- **What it shows:** Which payment methods (UPI, Credit Card, Cash on Delivery, etc.) are associated with higher churn.
- **Insight:** If COD customers churn more, it may indicate trust or delivery issues. If UPI users churn less, promoting digital payments could improve retention.

3. Churn by Tenure

- **What it shows:** How churn varies across customer lifecycle stages.
- **Insight:** New customers (low tenure) often churn quickly if onboarding is weak. Long-tenure customers churning may indicate dissatisfaction or competitors offering better deals. Tailor retention strategies by tenure group.

4. Churn by Satisfaction Score

- **What it shows:** Correlation between satisfaction ratings and churn.
- **Insight:** Low satisfaction scores strongly predict churn. This highlights the need for proactive customer support, faster complaint resolution, and personalized engagement for low-score customers.

5. Churn by ComplaintReceived

- **What it shows:** Whether customers who raised complaints are more likely to churn.
- **Insight:** If complaint-driven churn is high, it signals poor resolution processes. Strengthening customer service and faster complaint handling can directly reduce churn.

6. Churn by City Tier

- **What it shows:** Geographic churn patterns across Tier 1, 2, and 3 cities.
- **Insight:** Higher churn in Tier 2/3 cities may indicate delivery challenges, limited product variety, or weaker brand presence. This helps prioritize logistics and marketing investments regionally.

7. High-Risk Segments

- **What it shows:** Top combinations of payment mode, order category, and gender with the highest churn rates.
- **Insight:** Pinpoints vulnerable customer segments (e.g., male COD users buying fashion items). These segments should be targeted with retention campaigns like discounts, loyalty rewards, or personalized communication.

Deliverables

The project will deliver the following outputs:

1. Cleaned Dataset

- Missing values imputed (mean for numeric, mode for categorical).
- Outliers removed from WarehouseToHome.
- Standardized categorical values (payment modes, device types, order categories).
- Derived fields (ComplaintReceived, ChurnStatus) added.

2. SQL Scripts

- Data cleaning scripts (imputation, transformation, column renaming, dropping).

- Analysis queries for churn rate, payment mode, tenure, satisfaction, complaints, city tier, and high-risk segments.

The SQL queries and dashboard-ready views revealed clear churn drivers:

- **Payment Mode:** Cash on Delivery customers showed higher churn, highlighting trust and delivery challenges.
- **Tenure:** Early-stage customers churned more, while long-tenure churn indicated dissatisfaction.
- **Satisfaction Score:** Low scores strongly correlated with churn, emphasizing the importance of customer experience.
- **Complaints:** Customers who raised complaints were significantly more likely to churn, underscoring the need for robust resolution processes.
- **City Tier:** Tier 2 and Tier 3 cities exhibited higher churn, pointing to regional logistics and service gaps.
- **High-Risk Segments:** Specific combinations (e.g., male COD users in fashion categories) emerged as vulnerable groups requiring targeted retention strategies.
- Overall, the project provided actionable insights that can guide e-commerce enterprises in designing **lifecycle-based engagement, payment promotions, complaint resolution improvements, and region-specific strategies** to reduce churn and strengthen customer loyalty.

Future Work

While this project focused on descriptive churn analysis, several extensions can enhance its impact:

Predictive Modeling

- Build machine learning models (e.g., logistic regression, decision trees, random forests) to predict churn probability at the individual customer level.
- Use features such as tenure, satisfaction, complaints, and payment mode to generate churn risk scores.

Advanced Segmentation

- Apply clustering techniques (e.g., K-Means, DBSCAN) to identify hidden customer segments beyond predefined categories.
- Enable hyper-personalized retention campaigns.

RFM Analysis

- Incorporate Recency, Frequency, and Monetary value metrics to evaluate customer loyalty and spending behaviour.

- Combine RFM with churn flags for deeper insights.

Dashboard Enhancements

- Integrate predictive churn scores into Power BI/Tableau dashboards.
- Add interactive filters (city tier, payment mode, satisfaction) for real-time drilldowns.

A/B Testing of Retention Strategies

- Implement targeted interventions (cashback offers, loyalty programs, complaint resolution improvements).
- Measure effectiveness through controlled experiments.

Expansion to Indian E-Commerce Context

- Extend dataset with realistic attributes (UPI adoption, COD trust factors, regional delivery times).
- Align insights with Flipkart/Amazon-style customer behavior for practical applicability.

Conclusion

The **E-Commerce Customer Churn Analysis** project successfully demonstrated how structured data cleaning, transformation, and exploratory analysis can uncover meaningful insights into customer attrition. By imputing missing values, standardizing categorical fields, and creating derived attributes such as ComplaintReceived and ChurnStatus, the dataset was transformed into a reliable foundation for analysis.